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GRADES 7–10

EMERGENCY SUB PLANS PACK

for DIGITAL TECHNOLOGIES

★ *Keep learning on track—even when plans change.* ★

NO-PREP ACTIVITIES

READY-TO-USE RESOURCES

PERFECT FOR EMERGENCY LESSONS

GREAT FOR SUBSTITUTE TEACHERS

ENGAGING DIGITAL TECH TASKS

PRINT OR DIGITAL USE

YOUR *LIFESAVER* IN THE CLASSROOM

SAVE TIME | REDUCE STRESS | KEEP STUDENTS

MADE BY TEACHERS FOR TEACHERS

BEFORE YOU NEED IT

Set this up while you are well

FOR THE REGULAR TEACHER

15 MINUTES

You are reading this because you are **well**. Spend fifteen minutes now and this pack will run itself on a day when you are not.

The one job you have today

Print pages 3–6 of this pack and leave them in a clearly labeled folder where your substitute teacher will find them — top drawer, on the desk, or with the front office. Everything a substitute needs is in those four pages.

You do not need to print the lessons in advance. The substitute teacher chooses one on the day and photocopies it.

SETTING IT UP · FIFTEEN MINUTES, ONCE

1. **Print the whole pack once.** Put it in a display folder or bind it. Write your name and room number on the cover.
2. **Print pages 3–6 a second time** and clip them to the front. These are the pages a substitute teacher reads first.
3. **Tell your department head where it lives.** On a genuine emergency day you will not be answering your phone, so somebody else needs to know.
4. **Cross off the lessons you have already used.** There is a tracker on page 4. Two years of substitute days will not repeat if you keep it current.
5. **Optional: leave a class list** and a note about any students who need specific support, seating or a reader.

Why these lessons are unplugged

On a substitute day the laptop cart is often locked, the login server is down, or nobody knows the room password. Every lesson here runs on paper.

Each lesson also has an **If the computers are working** note, so the same sheet still works when technology cooperates.

Why they are not busy work

Every lesson targets a genuine Digital Technologies concept and produces marked evidence you can put in a folder.

If you are away for a week, running four of these in sequence is a defensible unit of work, not a holding pattern.

A word about the answer keys

Every answer key sits on the **last page of its lesson**. If you hand a lesson to a substitute teacher, hand them the answer key too — it is the single thing that lets a non-specialist mark with confidence and answer a student who says *“but why?”*

This is Volume Four

Volume 4 stands completely on its own and shares nothing with Volumes 1, 2 or 3. There is no binary conversion, no flowchart, no data cleaning, no truth table and no DNS here. Owning all four gives you forty substitute lessons that never repeat.

SUBSTITUTE TEACHER · START HERE

You do not need to know this subject

READ THIS FIRST

2 MINUTES

You have walked into a Digital Technologies class. You may have never taught the subject. That is completely fine — this pack was written for you.

Read this and you are ready

You do not need to know how to code. Every lesson is a paper worksheet with a full answer key. Your job is to hand it out, keep the room settled, and check work as students finish. You are not expected to teach the content.

DO THESE FOUR THINGS

1. **Pick one lesson** from the table on the next page. Any of them works for any class from Grade 7 to Grade 10. If you cannot decide, use Lesson 04.
2. **Photocopy the student pages only.** Each lesson tells you exactly which page numbers to copy. One copy per student.
3. **Keep the teacher page and the answer key.** Do not photocopy those.
4. **Read the teacher page once** before the bell. It takes about two minutes and includes the exact words you can say to start the lesson.

If a student says “we have done this”

Ask them to prove it by completing the extension section at the end. If they genuinely have, switch them to a different lesson from the table — the pack has ten, and they will not have done them all.

If you finish early

Every lesson has an extension task on its final student page. Point students there. There is also a discussion prompt on the teacher page you can run as a whole class for the last five minutes.

What to leave for the regular teacher

Collect the completed sheets. Fill in the handover note on the last page of this pack — it takes ninety seconds and tells the regular teacher which lesson you used, how far the class got, and anything that happened.

Leave both on the desk.

SUBSTITUTE TEACHER · STEP ONE

Choose your lesson

PICK ANY ONE

10 OPTIONS

Every lesson is standalone. There is no order and no prerequisite — pick whichever suits the class in front of you.

#	LESSON	WHAT IT COVERS	DIFFICULTY	MINS	STUDENT PAGES
01	Pictures Made of Numbers	Bitmaps, resolution and color depth	★★	60	7–9
02	Sound Made of Numbers	Sampling, bit depth and audio files	★★	55	12–14
03	Stacks, Queues and the Undo Button	Data structures in real systems	★★★	55	17–19
04	Reading the Logs	Digital forensics and evidence	★★★	60	22–24
05	Version Control by Hand	Tracking changes and collaboration	★★	55	27–29
06	Building a School Network	Network devices, topology and capacity	★★	60	32–34
07	Design for Everyone	Accessibility and inclusive design	★	55	37–39
08	The Algorithm Decided	Automated decisions and fairness	★★	55	42–44
09	The Smart Home Audit	Connected devices and their risks	★★	55	47–49
10	Will a Machine Take This Job?	Automation, work and the future	★	55	52–54

Easiest to supervise

07, 09 and 10. Judgment and argument with a definite conclusion waiting at the end. Students have opinions, so nobody gets stuck.

Best for a quiet, working class

01, 02 and 03. Puzzle-like with exact answers. Strong students will race; that is what the extensions are for.

Best for a restless class

04, 05 and 08. An investigation, a folder full of chaos and a scholarship to argue about. Noisier, but productively so.

Used-lesson tracker · for the regular teacher

LESSON	DATE USED	CLASS	NOTES
01 — Images as data			
02 — Sound as data			
03 — Stacks and queues			
04 — Digital forensics			
05 — Version control			
06 — Network design			
07 — Accessibility			
08 — Automated decisions			
09 — Smart devices			
10 — Automation and work			

SUBSTITUTE TEACHER · STEP TWO

The first ten minutes

WHAT TO SAY

SCRIPT

The first four minutes decide the next fifty. Here is a script that works even if you have never met this class.

SAY THIS AT THE DOOR

> *"Come in, sit down, and leave your bags off the desks. You will not need a computer today. Everything is on paper."*

THEN, ONCE THEY ARE SEATED

> *"Your teacher is away. My name is ______. We are doing a proper Digital Technologies lesson today, not a free period. There is a worksheet, it gets collected at the end, and your teacher will see it."*

Why that last sentence matters

The single most useful thing you can say is that the work will be seen by their regular teacher. It is also true — there is a handover note at the back of this pack.

THE SHAPE OF EVERY LESSON IN THIS PACK

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Settle, hand out sheets	Use the script above
5–10	Read the opening page together	Ask one student to read it aloud
10–45	Students work through the tasks	Circulate. Check answers against the key
45–55	Extension task for those who finish	Point them to the last student page
55–60	Collect sheets, pack up	Fill in the handover note

If a student refuses to work

Offer the smaller ask: *"Just do question one and show me."* Most refusals are about not knowing where to start, not defiance.

Note it on the handover sheet. Do not escalate a single lesson into a battle you will not be there to finish.

If the whole class claims they cannot do it

Do the first question together on the board. Every lesson has a worked example on the first student page.

If it is genuinely too hard for the grade level, switch to Lesson 07 or Lesson 10 — both need no prior knowledge at all.

Photocopying reminder

Student pages only. Each lesson's teacher page states the exact page range. The **answer key is always the last page** of the lesson — keep it.

LESSON 01 · TEACHER BRIEFING

Pictures Made of Numbers

READ BEFORE THE BELL

2 MINUTES

Students decode an image from its binary, encode one of their own, then calculate what a photograph actually costs in bytes and work out why enlarging one turns it into squares.

You do not need to know this subject

Every answer is a shaded grid or a multiplication. The completed images and every number are in the key.

Photocopy these pages only

Pages 7–9

One per student. **Do not photocopy this page or the answer key.**

What students need

A pen, and a pencil for shading squares.

WHAT TO SAY TO START

> *"A photograph contains no faces, no colors and no sky. It contains several million numbers. Today you find out exactly which numbers, and how many."*

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the pixel panel and the worked row	Shade the worked row together
14–28	Task 1.1 — decode and encode	Self-checking: a shape appears or it does not
28–42	Task 1.2 — color depth and file size	Arithmetic. Every answer is in the key
42–54	Task 1.3 — resolution, and why zooming fails	The vector question is the payoff
54–60	Tasks 1.4 and 1.5, collect sheets	Discussion prompt below

Questions students will ask

"Is 1 black or white?"
It is whatever both ends agreed. On this sheet 1 means shaded. Somebody has to decide, and that decision is part of the format.

"Why 24 bits for color?"
Eight bits each for red, green and blue. Three numbers per pixel, each from 0 to 255.

"My file size looks wrong."
Check they multiplied the pixels first, then the bits per pixel, then divided by 8 to get bytes.

If they finish early

Task 1.4 covers metadata. Strong students can then work out why an image posted publicly can reveal where it was taken, and what a platform does about that.

If the computers are working

Students can open any photo's properties and compare the stored file size with the number they calculated. The stored file is far smaller, which is a neat lead-in to why compression exists.

Curriculum focus · for the regular teacher's records

How digital systems represent images; pixels, resolution and color depth; calculating file sizes; bitmap and vector graphics; metadata.
Discussion prompt for the last five minutes: *"The screen is showing a face. Where is the face actually stored, and in what form?"*

LESSON 01 · PICTURES MADE OF NUMBERS

Every picture is a grid

UNDERSTAND

16 MIN

NAME

CLASS

DATE

One bit per pixel

A black and white image is a grid of squares. Each square gets one bit: **1** means shaded, **0** means blank. Read left to right, top to bottom, and the picture rebuilds itself.

A color image works identically, except each pixel needs more bits, because it has to say which of millions of colors it is.

Task 1.1 · Decode, then encode

DO

a · decode this

00111100
01000010
10100101
10000001
10100101
10011001
01000010
00111100

Shade every 1.

What is it? ____________

b · now encode

Shade a capital **T** in this grid, then write the eight binary rows beside it.

________ ________
________ ________
________ ________
________ ________

How many bits did your T take? ______

c. Both images are 8 by 8. Explain why they take exactly the same number of bits even though one is far more complicated. ____

Color costs bits

KEY SKILL

20 MIN

Task 1.2 · How many colors, and how big

CALCULATE

The number of colors a pixel can be is **2 multiplied by itself once per bit**.

BITS PER PIXEL	HOW MANY COLORS	WHAT IT IS USED FOR
1		Fax, simple line art
2		Very early screens
4		Old game consoles
8		GIF images, icons
24		Photographs on any modern screen

Now the file sizes. A pixel count multiplied by bits per pixel gives bits; divide by 8 for bytes.

#	THE IMAGE	WORKING	SIZE IN BYTES
a	100 × 100, 1 bit per pixel		
b	100 × 100, 8 bits per pixel		
c	100 × 100, 24 bits per pixel		
d	A 12 megapixel photo at 24 bits per pixel		

e. Your phone says the photo in **d** is about 3 MB. Your answer is far bigger. What has happened to the rest?

LESSON 01 · PICTURES MADE OF NUMBERS

Resolution, and what breaks

STRETCH

16 MIN

Task 1.3 · Why zooming turns it into squares

THINK

#	QUESTION	ANSWER
a	How many pixels in a 1920 × 1080 screen?	
b	You enlarge a 200 × 200 photo to fill it. How many pixels must be invented?	
c	Where does the computer get the colors for them?	
d	Why does the same enlargement not damage a school logo saved as a vector?	

Classify each statement **B** for bitmap or **V** for vector:

STATEMENT	B/V	STATEMENT	B/V
Stores a grid of pixel colors		Stores shapes, points and instructions	
Gets blurry when enlarged		The only sensible choice for a photograph	

Task 1.4 · What else is in the file

EXTENSION

An image file carries more than pixels. It also carries **metadata**: date and time, camera model, settings, and often the exact location.

a. Give one genuinely useful thing metadata makes possible: ______________________

b. A student posts a photo taken in their bedroom. What could the metadata reveal?

____

c. Most platforms strip metadata on upload. Why — and why would a professional photographer object?

____

____

Task 1.5 · The last word

THINK

The screen is showing a face. Explain in one sentence where the face is actually stored and what it is stored as.

____

____

Before you hand this in

CHECKPOINT

Check what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

____

____

Teacher initials ______________________

LESSON 01 · TEACHER COPY

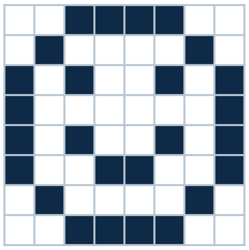
Answer key

Pictures Made of Numbers

NOT FOR STUDENTS

ANSWERS

Task 1.1 · the images



It is a face — a round outline, two eyes and a smile.

b. A capital T is `11111111` then rows of `00011000`. Any recognizable T is correct, provided the binary matches the shading exactly.

Both take 64 bits — 8 rows of 8. **c.** A bitmap stores one value per pixel whatever the picture shows, so blank space costs the same as detail. That is the weakness of the format, and the reason compression exists.

Task 1.2 · colors and sizes

Colors: 1 bit = 2 · 2 bits = 4 · 4 bits = 16 · 8 bits = 256 · 24 bits = **16,777,216**.

a. 10,000 × 1 ÷ 8 = **1,250 bytes**

b. 10,000 × 8 ÷ 8 = **10,000 bytes**

c. 10,000 × 24 ÷ 8 = **30,000 bytes**

d. 12,000,000 × 24 ÷ 8 = **36,000,000 bytes**, about 36 MB

e. The file was compressed. A JPEG discards detail the eye is poor at noticing, turning 36 MB into about 3 MB. Students who name the loss as well have the better answer.

Task 1.3 · resolution

a. 1920 × 1080 = **2,073,600** pixels.

b. The original has 40,000 pixels; the screen needs 2,073,600. Over **two million** pixels have to be invented.

c. It guesses, by copying or averaging neighboring pixels. Nothing new is discovered, which is why the result is blocky or smeared. The detail was never in the file.

d. A vector stores instructions — circle here, this radius, this color — so it is redrawn at whatever size is asked for and is always exactly sharp.

B: grid of pixel colors; blurry when enlarged. **V:** shapes and instructions; and the photograph is **B**, because no reasonable set of instructions describes a face.

Task 1.4 and 1.5

a. Sorting photos by date, finding where a holiday photo was taken, proving when a picture was made, or matching a photo to a place on a map.

b. The coordinates of their home, accurate to a few yards, plus the date, the time and the device — none of which they meant to publish.

c. Platforms strip it to reduce file size and to avoid publishing user locations by accident. A photographer objects because the metadata carries their name, their copyright notice and their settings, all of which vanish with it.

The last word: the face is stored as a long list of numbers — three per pixel — and it only becomes a face when a screen lights up in that pattern and a human brain interprets it. Nothing in the file resembles a face in any way.

LESSON 02 · TEACHER BRIEFING

Sound Made of Numbers

READ BEFORE THE BELL

2 MINUTES

Students turn a wave into whole numbers, calculate the true size of a three-minute song, then work out why a phone call sounds flat and why sampling too slowly produces the wrong note entirely.

You do not need to know this subject

The lesson is rounding and multiplication. Every value and every file size is worked out in the key.

Photocopy these pages only

Pages 12–14

One per student. **Do not photocopy this page or the answer key.**

What students need

A pen. A calculator is genuinely useful for Task 2.2.

WHAT TO SAY TO START

> *"Sound is a wave in the air, and a computer cannot store a wave. It can only store numbers. Today you find out how many numbers a song is, and what gets lost between the two."*

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–12	Read the sampling panel together	The measure-and-write-it-down idea does the work
12–26	Task 2.1 — round the samples	Exact answers. Grade as you circulate
26–40	Task 2.2 — the file size	Calculator allowed. Key has every step
40–50	Task 2.3 and 2.4 — what is thrown away, and sampling too slowly	Key has models
50–55	Task 2.5, collect sheets	Discussion prompt below

Questions students will ask

"Why divide by 8?"
The formula produces bits. Eight bits make a byte, and file sizes are quoted in bytes.

"Why is stereo twice as big?"
Two channels means two separate streams of numbers, one per ear.

"Why 44,100 samples a second?"
Because you need at least double the highest frequency, and human hearing stops around 20,000 Hz. That is Task 2.4.

If they finish early

Task 2.4 covers what happens when you sample too slowly. Strong students can then explain why the spokes of a wheel appear to spin backwards in a video, which is exactly the same effect.

If the computers are working

Any audio editor will show the waveform and let students export the same clip at different sample rates and bit depths. Listening to an 8-bit version makes the lesson permanent.

Curriculum focus · for the regular teacher's records

How digital systems represent sound; sampling rate, bit depth and quantisation; calculating file sizes; lossy audio compression. Discussion prompt for the last five minutes: *"A phone call and a music track both travel over the same internet. Why does one sound flat and the other does not?"*

Measuring a wave

UNDERSTAND

14 MIN

NAME

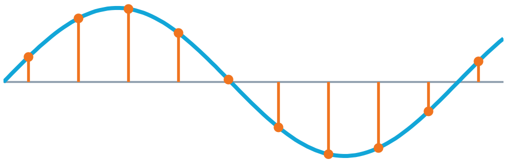
CLASS

DATE

Thousands of measurements a second

A microphone turns air pressure into a wave. The computer measures the height of that wave at regular instants and writes down a whole number each time.

Sample rate is how many measurements per second. **Bit depth** is how many different heights each measurement is allowed to be. Everything in between the measurements is gone forever.



Task 2.1 · Round the samples

CALCULATE

The wave was measured at ten instants. With **4 bits** each measurement may be any whole number from −8 to 7. With **3 bits** only even numbers are available: −8, −6, −4, −2, 0, 2, 4, 6.

SAMPLE	1	2	3	4	5	6	7	8	9	10
True height	0.0	3.1	5.4	6.2	4.8	1.2	−2.6	−5.9	−6.4	−3.3
4-bit value										
3-bit value										

a. Which sample loses the most accuracy at 3 bits? ______ How far off is it? ______

b. The difference between the true height and the stored number has a name: quantisation error. What would a listener hear if it were large? ____________________

c. Doubling the sample rate would change the table how? ______________________________

LESSON 02 · SOUND MADE OF NUMBERS

What a song weighs

KEY SKILL

18 MIN

Task 2.2 · The file size formula

CALCULATE

samples per second × bits per sample × channels × seconds ÷ 8 = bytes

#	THE RECORDING	WORKING	SIZE
a	CD quality: 44,100 samples, 16 bits, stereo, 3 minutes		
b	The same track in mono		
c	The same track at 22,050 samples per second, stereo		
d	A voice memo: 8,000 samples, 8 bits, mono, 1 minute		

e. A downloaded copy of track **a** is about 3 MB. Roughly how many times smaller is that than your answer?

______

f. Which single change in the table above saved the most space, and what did it cost?

____

Task 2.3 · What gets thrown away

JUDGE

A lossy audio format discards sound the ear is unlikely to notice. Mark each statement **true** or **false** and correct any that are wrong.

#	STATEMENT	T/F	IF FALSE, WHAT IS RIGHT
a	Sounds above the range of human hearing are removed		
b	A quiet sound just after a loud one may be removed		
c	The file can be turned back into the original exactly		
d	Compressing the same file twice makes it worse each time		
e	A higher bit rate means more of the sound is kept		

LESSON 02 · SOUND MADE OF NUMBERS

Too few measurements

STRETCH

16 MIN

Task 2.4 · Sampling too slowly

PREDICT

To record a sound properly you need **at least twice as many samples per second as the highest frequency in it**. Below that, the measurements fit a completely different, lower wave — and that is what plays back.

#	QUESTION	ANSWER
a	Human hearing reaches about 20,000 Hz. What is the minimum sample rate?	
b	CD audio uses 44,100. Why more than the minimum?	
c	A phone call uses 8,000 samples per second. What is the highest sound it can carry?	
d	Why does a voice still sound like that person on the phone?	

e. In a video, a spinning wheel sometimes appears to turn backwards. Explain how that is the same problem.

Task 2.5 · The last word

THINK

A phone call and a music track travel over the same internet, to the same phone, through the same speaker. Explain in one or two sentences why one sounds flat and the other does not.

Before you hand this in

CHECKPOINT

Check what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials

LESSON 02 · TEACHER COPY

Answer key

Sound Made of Numbers

NOT FOR STUDENTS

ANSWERS

Task 2.1 · the sampled values

SAMPLE	1	2	3	4	5	6	7	8	9	10
True height	0.0	3.1	5.4	6.2	4.8	1.2	−2.6	−5.9	−6.4	−3.3
4-bit	0	3	5	6	5	1	−3	−6	−6	−3
3-bit	0	4	6	6	4	2	−2	−6	−6	−4

a. Sample 3 (5.4 stored as 6) and sample 5 (4.8 stored as 4) are each 0.8 out; accept either. Sample 7 at 0.6 is the next closest. The 4-bit row is never more than 0.5 out, which is the whole argument for more bits.

b. A rough, gritty, hissing quality behind the sound — the error is heard as noise. An 8-bit recording sounds like an old computer game for exactly this reason.

c. Twice as many columns, each closer together in time. The values would not become more accurate — sample rate and bit depth fix different problems, and confusing the two is the most common misunderstanding here.

Task 2.2 · the file sizes

a. 44,100 × 16 × 2 × 180 ÷ 8 = **31,752,000 bytes**, about **31.8 MB**

b. Half of a: **15,876,000 bytes**, about 15.9 MB

c. Half of a: **15,876,000 bytes** — the same saving by a completely different route

d. 8,000 × 8 × 1 × 60 ÷ 8 = **480,000 bytes**, about 0.5 MB

e. About **ten times** smaller.

f. Both b and c saved the same amount. Mono costs you the stereo image — where instruments sit in the space. Halving the sample rate costs the highest frequencies, so cymbals and consonants go dull. Full credit for naming what was given up, not just the number.

Task 2.3 · what gets thrown away

a. True. There is no point storing what nobody can hear.

b. True. A loud sound briefly masks quieter ones near it in time and pitch, and the format exploits that.

c. False. Lossy means gone. Only a lossless format can rebuild the original exactly.

d. True. Each compression works on the previous approximation, so the damage accumulates.

e. True. More bits per second means less was discarded — and a bigger file.

The pattern worth naming: every one of these is a trade of quality for size, decided by somebody who assumed things about your ears and your speakers.

Task 2.4 · sampling too slowly

a. At least **40,000** samples per second.

b. Extra headroom so the filters that remove the very highest frequencies have room to work without damaging the sounds just below. 44,100 also had practical origins in the video equipment used to master early recordings.

c. About **4,000 Hz** — half the sample rate.

d. Almost everything that identifies a voice sits below 4,000 Hz. What is lost is the brightness above it, which is why *f* and *s* sounds are hard to tell apart on a call.

e. The camera samples 25 or 30 times a second. If the wheel turns slightly less than a full spoke-gap between frames, the spokes appear to have moved backwards. The measurements fit a slower motion that never happened — identical to a wave sampled too slowly.

Task 2.5 · the closing question

A phone call is deliberately recorded at a low sample rate and heavily compressed, because the priority is fitting many simultaneous calls into limited bandwidth with the smallest possible delay. Music is sampled at more than five times that rate, because the priority is sounding good and a few seconds of buffering costs nothing.

Accept any answer reaching ‘the call throws away the high frequencies on purpose’. The strongest answers mention that a call cannot wait for late data while a music stream can.

LESSON 03 · TEACHER BRIEFING

Stacks, Queues and the Undo Button

READ BEFORE THE BELL

2 MINUTES

Students trace the same eight operations through a stack and a queue, watch the answers diverge, then work out why the browser back button needs two of them and why undo eventually forgets.

You do not need to know this subject

Every answer is a short list of numbers, and both traces are written out step by step in the key.

Photocopy these pages only

Pages 17–19

One per student. **Do not photocopy this page or the answer key.**

What students need

A pen. Crossing out rather than rubbing out keeps the working visible.

WHAT TO SAY TO START

> *"There are two ways to deal with a pile of work: newest first, or oldest first. Both are everywhere in every computer you have used, and choosing the wrong one is a real bug."*

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–12	Read the two panels together	Plates and a ticket queue do all the work
12–26	Task 3.1 — trace the stack	Exact answers. Key has every step
26–38	Task 3.2 — the same operations on a queue	The difference is the lesson
38–50	Task 3.3 and 3.4 — which one, and the back button	Key has both
50–55	Task 3.5, collect sheets	Discussion prompt below

Questions students will ask

"Which end do I take from?"
A stack: the end you last added to. A queue: the opposite end. That single difference produces everything else.

"What does POP return?"
The item that was removed. Students often write the remaining list and forget to record what came off.

"Can a stack run out?"
Yes. Popping an empty stack is an error, and a stack that grows forever is a real crash with a real name: stack overflow.

If they finish early

Task 3.4 rebuilds the browser back and forward buttons from two stacks. Strong students can then explain exactly why visiting a new page clears the forward history, which almost nobody can articulate.

If the computers are working

Students can test the browser prediction in Task 3.4 directly: go back twice, click a new link, then look at what forward does. Predict first, then check.

Curriculum focus · for the regular teacher's records

Abstract data structures: stacks and queues; last-in-first-out and first-in-first-out behavior; choosing an appropriate structure for a problem.
Discussion prompt for the last five minutes: *"Undo remembers only your last twenty or so actions. Why not all of them?"*

Two ways to hold a pile of work

UNDERSTAND

14 MIN

NAME

CLASS

DATE

Stack · last in, first out

A pile of plates. You add to the top and take from the top, so the newest item leaves first.

PUSH adds. **POP** removes and hands back the top item.

Queue · first in, first out

A line at a counter. You join the back and are served from the front, so the oldest item leaves first.

ADD joins the back. **REMOVE** takes from the front.

Task 3.1 · Trace the stack

TRACE

The stack starts empty. Write the whole stack after each operation, with the top on the right.

#	OPERATION	THE STACK IS NOW	WHAT CAME OUT
1	PUSH 3		
2	PUSH 7		
3	PUSH 1		
4	POP		
5	PUSH 9		
6	POP		
7	POP		
8	PUSH 5		

a. In what order did the items come out? ______________________

b. One item was pushed and never popped. Which? ______

Same operations, different answers

KEY SKILL

18 MIN

Task 3.2 · Now as a queue

TRACE

Exactly the same eight operations, but this time the front of the queue is on the left and you remove from the front.

#	OPERATION	THE QUEUE IS NOW	WHAT CAME OUT
1	ADD 3		
2	ADD 7		
3	ADD 1		
4	REMOVE		
5	ADD 9		
6	REMOVE		
7	REMOVE		
8	ADD 5		

a. Order out this time: ____________________

b. One number left the stack second and the queue last. Which? _____

Task 3.3 · Which one would you use?

DECIDE

SITUATION	S/Q
Undo in a word processor	
Printing jobs on a shared printer	
The browser back button	
A support call waiting line	

SITUATION	S/Q
Checking that every bracket is closed	
Messages waiting to be delivered	
Reversing a word letter by letter	
Students waiting in line at the cafeteria	

One of these eight would cause a genuine argument if you chose the wrong one. Which, and what would people notice? ______________________________

Back, forward, and forgetting

STRETCH

16 MIN

Task 3.4 · The browser needs two stacks

PREDICT

A browser keeps a **back** stack and a **forward** stack. Visiting a page pushes the current one onto back. Pressing back moves a page from back to forward, and the other way round.

#	WHAT YOU DO	BACK STACK	FORWARD STACK	PAGE YOU ARE ON
1	Visit A, then B, then C			
2	Press back			
3	Press back			
4	Press forward			
5	Click a link to page D			

a. After step 5, what happens if you press forward? ______________________

b. Explain why the browser has to do that — what would forward even mean?

____

Task 3.5 · The last word

THINK

Undo remembers roughly your last twenty actions and no more. Give the real reason, and say what would have to be stored for unlimited undo.

____

____

____

Before you hand this in

CHECKPOINT

Check what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

____

____

Teacher initials ______________________

Answer key

Stacks, Queues and the Undo Button

NOT FOR STUDENTS

ANSWERS

Task 3.1 · the stack

#	STACK	OUT
1	3	
2	3 7	
3	3 7 1	
4	3 7	**1**
5	3 7 9	
6	3 7	**9**
7	3	**7**
8	3 5	

a. Out in the order **1, 9, 7**. **b.** The 3 — pushed first, still sitting at the bottom at the end. Nothing on top of it can ever be reached past it.

Task 3.2 · the queue

#	QUEUE	OUT
1	3	
2	3 7	
3	3 7 1	
4	7 1	**3**
5	7 1 9	
6	1 9	**7**
7	9	**1**
8	9 5	

a. Out in the order **3, 7, 1** — exactly the order they arrived.

b. The **9**: second out of the stack, and still waiting at the end of the queue. Same eight instructions, completely different behavior, and neither structure is wrong — they answer different questions.

Task 3.3 · which structure

- **Undo** — stack. Undo the most recent thing first
- **Printing jobs** — queue. First submitted, first printed
- **Back button** — stack. The most recent page is the one behind you
- **Support calls** — queue. Anything else is unfair
- **Matching brackets** — stack. Each closing bracket must match the most recently opened one
- **Messages waiting** — queue. Delivery order should match sending order
- **Reversing a word** — stack. Push every letter, pop them all
- **Cafeteria line** — queue, obviously

The argument would be the support calls or the cafeteria. A stack serves the newest arrival first, so the person who has waited longest is served last — and the earliest callers might never be served at all. People notice queue-jumping within seconds, which is a useful reminder that a data structure choice is sometimes an ethical one.

Task 3.4 · the two stacks

#	BACK	FORWARD	ON
1	A B	empty	C
2	A	C	B
3	empty	C B	A
4	A	C	B
5	A B	**cleared**	D

a. Nothing. The forward button is grayed out.

b. Forward means ‘the page I was on before I pressed back’. Once you go somewhere new from here, that future never happened — there is no longer any page it could sensibly take you to. Every browser does this and almost nobody can say why until they have drawn the two stacks.

Task 3.5 · the closing question

Every undo step has to store enough information to reverse the change — the deleted text, the old formatting, the previous state of the image. That is memory, and it is held for as long as the document is open.

Unlimited undo would mean keeping every version of everything since the file was opened, which for an image or video editor is gigabytes within minutes. The limit is a deliberate trade of memory against regret.

Accept any answer reaching ‘each step costs memory’. The best answers note that professional editors solve it by writing the history to disk instead, which is exactly how version control begins.

LESSON 04 · TEACHER BRIEFING

Reading the Logs

READ BEFORE THE BELL

2 MINUTES

Students investigate a real-shaped server log, build the timeline of an intrusion, then decide which accusations the evidence actually supports. The most gripping lesson in the volume.

You do not need to know this subject

Everything needed is in the log on the student page. The answers are times, filenames and yes or no, and all of them are in the key.

Photocopy these pages only

Pages 22–24

One per student. **Do not photocopy this page or the answer key.**

What students need

A pen. A highlighter is genuinely useful here.

WHAT TO SAY TO START

> *"Every system keeps a record of what happened, and almost nobody reads it until something goes wrong. Today you are the person who reads it."*

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–12	Read the log together	Read three lines aloud so the format is clear
12–28	Task 4.1 — find the unusual session	Exact answers. Key has all
28–40	Task 4.2 — build the timeline	Ordering. One correct sequence
40–52	Task 4.3 — what it proves	The best discussion in the volume
52–60	Tasks 4.4 and 4.5, collect sheets	Discussion prompt below

Questions students will ask

"So Jordan did it?"
That is Task 4.3 and the answer is more careful than it looks. Do not settle it early.

"What is an IP address doing here?"
It says roughly where the connection came from. Two of them are inside the school and one is not, which is the thread to pull.

"Why does the time matter so much?"
Because order is what turns separate facts into a story. Task 4.2 is the whole point of the lesson.

If they finish early

Task 4.4 covers evidence handling. Strong students can then explain why an investigator makes a copy of a drive before examining it, and why the copy is checked against the original.

If the computers are working

Most operating systems keep a login and event log a student can open and read on a school device, with permission. Even five lines of a real log makes the format concrete.

Curriculum focus · for the regular teacher's records

Data security incidents; interpreting system logs and audit trails; evidence, inference and the limits of what data can prove. Discussion prompt for the last five minutes: *"The log proves an account did something. What would it take to prove a person did?"*

LESSON 04 · READING THE LOGS

The record nobody reads

UNDERSTAND

16 MIN

NAME	CLASS	DATE

The incident

A school office server holds staff files. On Friday morning a payroll spreadsheet turned up on a public file-sharing site. Below is the server log for the previous day. Addresses beginning `10.` are inside the school; anything else is outside it.

server.log · Thursday, August 14 AUDIT TRAIL

```
07:58  j.patel  LOGIN     10.2.14.6      ok
08:31  j.patel  OPEN      /reports/term3.xlsx
09:02  a.novak  LOGIN     10.2.14.9      ok
09:05  a.novak  OPEN      /staff/schedule.docx
12:44  j.patel  LOGOUT
15:10  a.novak  LOGOUT
18:55  MAILGW   j.patel opened link in "Payroll update required"
19:12  j.patel  LOGIN     203.0.113.44   ok
19:13  j.patel  OPEN      /admin/salaries.xlsx
19:15  j.patel  DOWNLOAD  /admin/salaries.xlsx
19:18  j.patel  OPEN      /admin/contracts/
19:22  j.patel  DOWNLOAD  /admin/staff_list.csv
19:26  j.patel  LOGIN     203.0.113.44   FAILED (admin console) x3
19:31  j.patel  LOGOUT
```

Task 4.1 · Read the evidence

INVESTIGATE

#	QUESTION	ANSWER
a	Which login is unusual, and what makes it unusual?	
b	How long did that session last?	
c	Which two files were taken?	
d	What was attempted at 19:26, and did it work?	
e	What happened 17 minutes before the unusual login?	

Turning facts into a story

KEY SKILL

18 MIN

Task 4.2 · Build the timeline

ORDER

Number these 1 to 7 in the order they must have happened, then write the one-sentence summary an investigator would put at the top of the report.

ORDER	EVENT
	Files were downloaded from the admin folder
	A convincing email was sent to a staff member
	Someone tried to reach the admin console and failed
	The staff member opened the link and entered their details
	The spreadsheet appeared on a public sharing site
	Someone logged in as that staff member from outside the school
	The attacker logged out

The summary:

a. Which single event, if it had not happened, would have stopped all the others?

Task 4.3 · What does it actually prove?

JUDGE

Mark each claim **supported**, **not supported** or **needs more evidence**, and say why.

#	THE CLAIM	VERDICT	WHY
a	Jordan Patel stole the salary file		
b	Jordan Patel’s account was used to download the file		
c	The attacker was outside the school building		
d	The attacker did not have the admin password		
e	Jordan Patel was careless		

Claims a and b look almost identical. Explain the difference in one sentence, and why it matters enormously.

LESSON 04 · READING THE LOGS

Handling the evidence

STRETCH

16 MIN

Task 4.4 · Do not touch the machine

ORDER

Number these steps 1 to 5 in the order a professional would carry them out.

ORDER	STEP
	Examine the copy, never the original
	Record who has held the evidence, and when
	Make an exact copy of the drive
	Check that the copy matches the original exactly
	Isolate the machine and leave it as it is

a. A teacher says “I had a quick look through the folder to see what was taken.” What has that cost the investigation? ____________________

b. Why check that the copy matches the original? What would a defense lawyer say if nobody did?

____

Task 4.5 · The last word

THINK

The attacker could have deleted the log. Explain whether that would have been good news or bad news for the investigator — and say what a school should do so that deleting it would not help.

____

____

____

Before you hand this in

CHECKPOINT

Check what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

____

____

Teacher initials ____________________

Answer key

Reading the Logs

NOT FOR STUDENTS

ANSWERS

Task 4.1 · reading the evidence

a. 19:12 — the same account, but from `203.0.113.44`, which is outside the school, and at an hour when the account had already logged out at 12:44.

b. 19:12 to 19:31 — **19 minutes**.

c. `/admin/salaries.xlsx` and `/admin/staff_list.csv`.

d. Three failed attempts to reach the admin console. They failed, which tells you the attacker had the staff password but not an administrator one.

e. At 18:55 the mail gateway recorded that j.patel opened a link in an email titled 'Payroll update required'. That is where the password was handed over, and it is the first event in the story.

Task 4.2 · the timeline

1 A convincing email was sent · **2** The staff member opened the link and entered their details · **3** Someone logged in from outside the school · **4** Files were downloaded · **5** Someone tried the admin console and failed · **6** The attacker logged out · **7** The spreadsheet appeared publicly.

Model summary: a staff account was taken over through a phishing email at 18:55 and used from an external address between 19:12 and 19:31 to download two files from the admin folder.

a. The link being opened. Every later event depended on the attacker having a working password, and no technical control on the server was defeated at any point. Accept 'the email getting through' with reasoning.

Task 4.3 · what the log proves

#	VERDICT	WHY
a	**Not supported**	The log records an account, not a person. Somebody else had the password by 19:12
b	**Supported**	This is exactly and only what the log records
c	**Needs more evidence**	The address is outside the school network. It could still be someone sitting in the parking lot, or a service hiding the real location
d	**Supported**	Three failed attempts at the admin console is good evidence they did not have it
e	**Needs more evidence**	They opened a convincing phishing email. Whether that is carelessness depends on how good the email was, and most are very good

The difference between a and b is the entire discipline. A log records what an *account* did. Attaching a person to that account needs other evidence — where they were, what device was used, whether anyone else could have known the password. Investigations that skip this step accuse the victim of the phishing attack rather than the attacker, and it happens regularly.

Task 4.4 · handling the evidence

1 Isolate the machine and leave it as it is · **2** Make an exact copy of the drive · **3** Check the copy matches the original · **4** Examine the copy · **5** Record who has held the evidence throughout.

Step 5 runs alongside all of them — accept it anywhere provided the copy is made before the examining.

a. Opening files changes the record: last-accessed times are rewritten, and there is now no way to distinguish the teacher's looking from the attacker's. Evidence has been destroyed by somebody trying to help, which is how it is usually destroyed.

b. The check proves the copy is identical to what was seized. Without it, a defense lawyer argues the file could have been altered at any point after seizure — and nobody could show otherwise.

Task 4.5 · the closing question

Deleting the log is **bad news for the attacker**. It rarely erases everything, it is itself recorded, and an unexplained gap in a log is strong evidence that something happened in it. An attacker who cleans up has told you exactly which window to investigate.

The defense is to send logs somewhere the attacker cannot reach: a separate machine, written once, that only accepts new entries and never edits. Accept any answer reaching 'keep the logs somewhere else'.

LESSON 05 · TEACHER BRIEFING

Version Control by Hand

READ BEFORE THE BELL

2 MINUTES

Students face a folder containing eight versions of the same document, work out which is current and why nobody can tell, then resolve a conflict a computer cannot resolve for them.

You do not need to know this subject

No software is involved. Every answer is a judgment about filenames and text, and the key gives a model for each.

Photocopy these pages only

Pages 27–29

One per student. **Do not photocopy this page or the answer key.**

What students need

A pen. Nothing else.

WHAT TO SAY TO START

> *"Everyone in this room has a folder with a file called final in it, and another called final two. Today you find out what professionals do instead, and why it is not just tidiness."*

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the folder together	Ask who recognizes it. Every hand goes up
14–28	Task 5.1 — which one is current	The honest answer is that nobody knows
28–40	Task 5.2 — write the change messages	Judgment. Key has models
40–52	Task 5.3 — the conflict	The hardest and best task on the sheet
52–55	Tasks 5.4 and 5.5, collect sheets	Discussion prompt below

Questions students will ask

"Why not just use the date the file was saved?"
Opening a file can change it. Copying it definitely does. Saved dates lie more often than students expect.

"Why write the date backwards?"
So that sorting by name also sorts by date. That is the entire trick and it works everywhere.

"Can the computer merge it for me?"
Sometimes. Task 5.3 is about exactly when it cannot, and why.

If they finish early

Task 5.4 introduces branching. Strong students can then explain why you would branch before trying something risky rather than after, and what that costs.

If the computers are working

Any document tool with version history shows the same idea working properly. Comparing two versions side by side makes Task 5.3 concrete in a way paper cannot.

Curriculum focus · for the regular teacher's records

Managing and documenting the development of digital solutions; version control, collaboration and change management; working in teams on shared files. Discussion prompt for the last five minutes: *"Version control means you can always go back. Why does that change how people work, not just what they can recover?"*

LESSON 05 · VERSION CONTROL BY HAND

The folder everybody recognizes

UNDERSTAND

14 MIN

NAME	CLASS	DATE

Eight files, one document

This is a real shared folder for a group assignment due on Friday. Every file is a version of the same report. Three students have been editing it.

GroupAssignment/ SHARED DRIVE

```
report.docx
report_v2.docx
report_final.docx
report_final_v2.docx
report_final(1).docx
report_FINAL_use_this_one.docx
report_janedits.docx
report_final_v2_MARK.docx
```

Task 5.1 · Which one is current?

INVESTIGATE

#	QUESTION	ANSWER
a	Which file would you open first, and why?	
b	How confident are you, out of 10?	
c	What does `report_final(1).docx` most likely mean?	
d	Two files could each be the newest. Which two?	
e	What three pieces of information are missing from every single name?	

Now rewrite three of them using a convention that sorts correctly and answers **e**:

LESSON 05 · VERSION CONTROL BY HAND

Saying what changed

KEY SKILL

18 MIN

Task 5.2 · Write the change message

WRITE

A saved version should carry a one-line message saying **what changed and why**. Rewrite each useless message.

#	WHAT WAS ACTUALLY DONE	THE MESSAGE THEY WROTE	A USEFUL MESSAGE
a	Fixed three spelling errors in the introduction	update	
b	Rewrote the conclusion after teacher feedback	asdf	
c	Added the survey results table to section 4	changes	
d	Deleted the whole methodology section by accident	final version	
e	Reformatted every heading to match the template	fixed it	

Which of these five would be most dangerous to lose track of, and why? ______________________

Task 5.3 · Two people, one paragraph

RESOLVE

Mia and Sam both edited the same sentence at the same time, from the same starting version.

Mia's version

The survey showed that 62% of students used a device after 10pm, which suggests a link with reported tiredness.

Sam's version

62% of students used a device after 10pm. We cannot claim this caused tiredness from a survey alone.

#	QUESTION	ANSWER
a	Can a computer merge these automatically?	
b	If Mia saved last, what happens to Sam's work?	
c	Who should decide which survives?	

d. Write a merged sentence that keeps what each of them was right about:

____

____

Trying something risky

STRETCH

16 MIN

Task 5.4 · Branching

EXTENSION

A **branch** is a copy of the whole project that you can wreck freely. If it works you merge it back; if it does not you delete it and the original was never touched.

#	QUESTION	ANSWER
a	You want to try a completely different layout. Branch before or after?	
b	Your teammate keeps working on the main version meanwhile. Is that a problem?	
c	Your experiment fails. What do you do?	
d	It works, but the main version has changed underneath you. Now what?	

e. Name one cost of branching. It is not free. ______________________

Task 5.5 · The last word

THINK

When every version is kept and nothing can be permanently lost, people work differently — not just more safely. Explain in one or two sentences what changes.

____

____

____

Before you hand this in

CHECKPOINT

Check what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

____

____

Teacher initials ______________________

LESSON 05 · TEACHER COPY

Answer key

Version Control by Hand

NOT FOR STUDENTS

ANSWERS

Task 5.1 · which is current

a. Most students choose `report_FINAL_use_this_one.docx` because it shouts loudest. That is not evidence.

b. Anything above 5 is overconfident. The honest answer is 2 or 3.

c. A duplicate created automatically when somebody downloaded or copied a file that already existed. It is usually an *older* copy wearing a newer name.

d. `report_FINAL_use_this_one` and `report_final_v2_MARK` — one claims authority, the other claims a later version number, and nothing distinguishes them.

e. **When** it was saved, **who** saved it, and **what changed**.

Model rename: `2026-08-14_report_v03_mia.docx`. The date first, written year-month-day, so sorting by name sorts by age — which is the point of writing it backwards.

Task 5.2 · useful messages

a. ‘Fixed three spelling errors in the introduction’

b. ‘Rewrote conclusion following teacher feedback on 12 Aug’

c. ‘Added survey results table to section 4’

d. ‘Methodology section removed — accidental, needs restoring’

e. ‘Reformatted all headings to match the school template’

d is the dangerous one. Labeled ‘final version’, a whole section has silently disappeared, and the message actively discourages anyone from looking. A message that hides a destructive change is worse than no message at all.

The test worth teaching: could somebody choose whether to open this version without opening it?

Task 5.3 · the conflict

a. No. Merging tools work line by line. Two different rewrites of the same sentence cannot both survive, and no software can judge which claim is better supported.

b. Sam’s version is overwritten and, in a plain shared folder, is simply gone. A real version control system refuses the save and reports a conflict instead.

c. The two authors, together. This is a disagreement about what the data shows, not a technical problem, and the tool’s job is only to make sure it is noticed.

d. Model merge: *62% of students used a device after 10pm. Reported tiredness was higher in this group, though a survey cannot show that one caused the other.* Sam is right about causation; Mia is right that the link is worth reporting.

Full credit requires keeping the number, keeping the observed link, and refusing to claim cause.

Task 5.4 and 5.5

a. Before. Branching after you have started means the risky work is already tangled with the version everyone depends on.

b. No — that is the entire point. Both lines of work continue independently.

c. Delete the branch. Nothing was risked and nobody else was affected.

d. Bring the main version’s changes into your branch, resolve anything that clashes, then merge back. Doing it in that order means the clash is your problem, not everyone’s.

e. The longer a branch lives, the further it drifts from the main version and the more painful the merge. Branches are cheap to make and expensive to leave lying around.

The last word: people take bigger risks. If any experiment can be abandoned without cost, you try the idea instead of arguing about it — and delete the ones that fail without anyone having to defend them.

LESSON 06 · TEACHER BRIEFING

Building a School Network

READ BEFORE THE BELL

2 MINUTES

Students follow a request out of a laptop and through six devices, compare three ways of wiring a building, then work out whether the school's connection can survive a grade level all streaming video at once.

You do not need to know this subject

Everything is matching, ordering and one multiplication. The device list and every number are in the key.

Photocopy these pages only

Pages 32–34

One per student. **Do not photocopy this page or the answer key.**

What students need

A pen. A calculator is useful for Task 6.3 but the numbers are round.

WHAT TO SAY TO START

> *"There is a moment every day when six hundred people press play at the same time. Today you find out exactly what has to happen for that to work, and what happens when it does not."*

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the six devices together	Ask which ones they have seen in the building
14–28	Task 6.1 — match and order	Exact answers. Key has both
28–40	Task 6.2 — three ways to wire a building	Comparison table. Key has it filled
40–52	Task 6.3 — does it fit?	The arithmetic is the payoff
52–60	Tasks 6.4 and 6.5, collect sheets	Discussion prompt below

Questions students will ask

"What is the difference between a switch and a router?"
A switch moves traffic inside one network. A router moves it between networks. That distinction is Task 6.1.

"Is wifi slower than cable?"
Usually, and more importantly it is shared. Thirty people on one access point divide what one cable would have given a single machine.

"Why is the internet slow at 11am?"
Because everybody is on it. Task 6.3 turns that into a number.

If they finish early

Task 6.4 tackles ninety students in a hall on one access point. Strong students can then explain why adding a second access point on the same channel does not help.

If the computers are working

A speed test on the school network at two different times of day produces two very different numbers, and the difference is the lesson. Ask students to predict which will be worse.

Curriculum focus · for the regular teacher's records

Networks: hardware components, topologies and transmission; bandwidth and network performance; designing a network to meet requirements. Discussion prompt for the last five minutes: *"Somebody says the internet is slow. Name three completely different places the problem could be."*

What a request passes through

UNDERSTAND

16 MIN

NAME

CLASS

DATE

Six devices, six jobs

Every network in every building is built from the same small set of parts. Knowing which one does what is most of what it takes to work out where a fault is.

Task 6.1 · Match the device to the job

CLASSIFY

DEVICE	ITS JOB
Switch	
Router	
Access point	

DEVICE	ITS JOB
Firewall	
Server	
Modem / ONT	

Now number these 1 to 6 in the order a request from a student laptop passes through them on its way to a website.

ORDER	DEVICE	ORDER	DEVICE
	The school's router		A wireless access point in the ceiling
	The firewall		The modem or ONT at the building edge
	A switch in the network closet		The laptop's own wireless adapter

Which one of the six is the only one a home network can often do without? ______________________

Wiring, and whether it fits

KEY SKILL
20 MIN

Task 6.2 · Three ways to wire a building

COMPARE

	STAR — EVERYTHING TO A CENTRAL SWITCH	RING — EACH DEVICE TO THE NEXT	MESH — MANY DEVICES LINKED TO MANY
Cable needed			
What happens if one cable fails			
Cost			
Easy to add a device?			

a. Which does your school almost certainly use, and why? ______

b. A star has one obvious weakness. What is it, and how would you reduce it?

______

Task 6.3 · Does it fit down the pipe?

CALCULATE

The school’s connection to the internet carries **100 megabits per second**. Streaming video needs about **5 megabits per second** per person.

#	QUESTION	WORKING	ANSWER
a	How many students can stream at once before it is full?		
b	A grade level of 150 all press play. What happens?		
c	The connection is upgraded to 500 Mbps. How many now?		
d	600 students each loading a 2 MB page once a minute. Does that fit?		

e. A video call and a large download both use the network. One is ruined by a delay of half a second and the other is not. Which, and why does that change how a network should treat them?

______

Ninety students in a hall

STRETCH

16 MIN

Task 6.4 · The wifi problem

DESIGN

The hall has **one** access point. Ninety students arrive with devices for an assembly quiz. Almost nothing loads.

#	QUESTION	YOUR ANSWER
a	Why does one access point struggle with ninety devices?	
b	Would a faster internet connection fix it? Why or why not?	
c	Give two fixes that would genuinely help	

d. A second access point is installed right beside the first, using the same channel. Predict what happens.

Task 6.5 · The last word

THINK

Somebody says “the internet is slow”. Name three completely different places the problem could be, and one question you would ask to tell them apart.

1.
2.
3.

Before you hand this in

CHECKPOINT

Check what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials

Answer key

Building a School Network

NOT FOR STUDENTS

ANSWERS

Task 6.1 · the devices

Switch — moves traffic between devices *inside* one network, sending each frame only to the port it needs.

Router — moves traffic *between* networks and decides where to send it next.

Access point — joins wireless devices to the wired network.

Firewall — allows or blocks traffic according to rules.

Server — stores files or runs services other machines request.

Modem / ONT — converts between the school's network and whatever the provider runs down the street.

Order: 1 wireless adapter · 2 access point · 3 switch · 4 firewall · 5 router · 6 modem. Accept firewall and router swapped — in most schools they are the same box, which is worth mentioning.

Without: a server. Most homes have none, and the switch is usually built into the router as well.

Task 6.2 · three topologies

	STAR	RING	MESH
Cable	Most — one run per device	Least	Far more than a star
One cable fails	That device only	Can break the whole ring	Traffic reroutes; nothing is lost
Cost	Moderate	Low	High
Adding a device	Easy	Awkward — break the ring	Easy

a. Star, in almost every building. A fault affects one device, adding a room is a cable run, and faults are easy to find.

b. The central switch is a single point of failure — lose it and the whole floor goes down. Reduce it with a second switch, a spare on the shelf, or a backup link between cupboards. Accept any answer that adds a second path.

Task 6.3 · the arithmetic

a. 100 ÷ 5 = **20 students**. Genuinely alarming for a school of hundreds.

b. It does not stop; it degrades. Everybody shares what there is, so video quality drops, buffering starts, and every other request queues behind it.

c. 500 ÷ 5 = **100 students**.

d. 600 × 2 MB = 1,200 MB per minute = 20 MB per second. One byte is 8 bits, so that is **160 Mbps** — more than the 100 Mbps link. It does **not** fit, and web pages are supposedly the light option.

e. The video call. A frame that arrives late is useless, whereas a download simply finishes a moment later. Networks handle this by prioritizing traffic that cannot wait — the same idea as an ambulance in traffic.

Task 6.4 · the hall

a. Wifi is shared airtime, not a private line. Only one device transmits at a time on a channel, so ninety devices take turns and each waits for the other eighty-nine.

b. No. The bottleneck is the air in the room, not the connection to the outside world. This is the most commonly bought wrong solution in schools.

c. Several access points spread through the hall on *different* channels · limit what the activity needs to download · stagger the start · use a quiz that loads once and runs locally.

d. Little or no improvement. Two access points on the same channel are competing for the same airtime, and now they interfere with each other as well. Channels must be different for the capacity to add up.

Task 6.5 · the closing question

Three genuinely different places: **the device** (an old machine, too many tabs, weak signal) · **the school network** (a saturated access point, a failing switch, one class downloading a large file) · **outside the school** (the internet connection is full, or the website itself is slow for everybody).

Good diagnostic questions: is it slow for the person next to you? Is it slow on every site or just one? Is it slow on a cabled machine? Each answer eliminates a whole layer, which is exactly how professionals work — and it is the reason the lesson lists the devices in order.

LESSON 07 · TEACHER BRIEFING

Design for Everyone

READ BEFORE THE BELL

2 MINUTES

Students identify who is shut out by eight ordinary design choices, write alt text for four images, then discover that most accessibility features were built for a minority and are now used by nearly everybody.

You do not need to know this subject

No technology and no specialist knowledge. Every answer is a judgment about who is excluded, and the key names them.

Photocopy these pages only

Pages 37–39

One per student. **Do not photocopy this page or the answer key.**

What students need

A pen. Nothing else.

WHAT TO SAY TO START

> *“Every design decision decides who can use the thing and who cannot. Most of the time nobody notices, because the people who cannot use it simply go away.”*

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–12	Read the three kinds of barrier together	The broken arm example does the work
12–28	Task 7.1 — the barrier audit	Eight answers, all in the key
28–38	Task 7.2 — color alone	Quick, and the fix is the point
38–50	Task 7.3 — write the alt text	One of the four is a trap
50–55	Tasks 7.4 and 7.5, collect sheets	Discussion prompt below

Questions students will ask

“How many people does this really affect?”
Around one person in five has a disability, and everybody is temporarily impaired sometimes. The panel makes that point before the tasks do.

“Is alt text just describing the picture?”
It is describing what the picture is *for*. Those are different, and Task 7.3 shows why.

“What if it makes it uglier?”
Good question, and mostly a false choice. Ask which of the eight fixes in Task 7.1 would damage how anything looks.

If they finish early

Task 7.4 covers using an interface without a mouse. Strong students can then work out why a pop-up that cannot be closed by keyboard is worse than one that cannot be closed at all.

If the computers are working

Students can navigate a real website using only the Tab key and see how far they get. Turning on a screen reader for two minutes is more persuasive than any explanation, if the school permits it.

Curriculum focus · for the regular teacher’s records

User-centered and inclusive design; accessibility requirements for digital solutions; evaluating designs against the needs of diverse users. Discussion prompt for the last five minutes: *“Captions were built for deaf viewers. Most people who use them now are not deaf. What does that tell you about designing for the edges?”*

LESSON 07 · DESIGN FOR EVERYONE

Three kinds of barrier

UNDERSTAND

14 MIN

NAME

CLASS

DATE

Permanent

One arm. Blind. Deaf. Around one person in five has a disability of some kind.

Temporary

A broken arm, an ear infection, an eye injury. Everybody, eventually.

Situational

Holding a baby. Bright sun on the screen. A noisy bus with no headphones.

The same design serves all three

A caption helps a deaf viewer, a person with an ear infection, and a student watching in a silent library. Designing for the hardest case usually improves it for everybody — which is why accessibility is a design skill and not a charity.

Task 7.1 · The barrier audit

DIAGNOSE

#	THE DESIGN CHOICE	WHO CANNOT USE IT, AND WHY
a	Errors are shown by turning the box red	
b	Video plays with no captions	
c	Light gray text on a white background	
d	The form clears itself after 60 seconds	
e	The only way to submit is to drag an icon	
f	Images have no text description	
g	Instructions are given in a video only	
h	Buttons are 8 pixels apart on a phone	

Which of these eight would cost the most to fix *after* the product is built? ____________________

Color, and describing pictures

KEY SKILL

20 MIN

Task 7.2 · When color is the only clue

REDESIGN

A dashboard shows the status of every school printer using color alone: green working, amber low on toner, red broken. About one boy in twelve cannot reliably tell red from green.

STATUS	WHAT IS SHOWN NOW	A SECOND CLUE YOU WOULD ADD
Working	Green dot	
Low on toner	Amber dot	
Broken	Red dot	

a. Your fix must survive being printed in black and white. Does it? ____________

b. Rank these three text combinations from easiest to hardest to read, and say why: black on white · mid-gray on white · yellow on white. ____

Task 7.3 · Write the alt text

WRITE

Alt text is read aloud by a screen reader. It should say what the image is **for**, not everything that is in it.

#	THE IMAGE	YOUR ALT TEXT
a	A graph showing enrollments rising from 800 to 1,100 since 2020	
b	A photo of the school gate, used to show visitors where to enter	
c	A decorative swirl at the top of the page	
d	A screenshot of an error message reading ‘file too large’	

One of these four should have **empty** alt text. Which, and why is empty better than a description?

____

LESSON 07 · DESIGN FOR EVERYONE

Without a mouse

STRETCH

16 MIN

Task 7.4 · Keyboard only

PREDICT

Some people cannot use a mouse at all. Everything must be reachable by pressing Tab to move and Enter to activate.

#	QUESTION	ANSWER
a	A form has Name, Email, Message, Send. What order should Tab visit them?	
b	A pop-up appears and its close button cannot be reached by Tab. What now?	
c	Why must the item you have tabbed to be visibly outlined?	
d	A menu only opens when the mouse hovers over it. What is the fix?	

Task 7.5 · The last word

THINK

Captions were built for deaf viewers, and most people using them now are not deaf. Explain in one or two sentences what that tells you about designing for the people at the edges.

Before you hand this in

CHECKPOINT

Check what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials

Answer key

Design for Everyone

NOT FOR STUDENTS

ANSWERS

Task 7.1 · the barrier audit

#	WHO IS EXCLUDED	THE FIX
a	Anyone color-blind, and anyone who now knows something is wrong but not what	Add a message naming the field and the problem
b	Deaf viewers, anyone in a noisy or silent place, anyone with weak English	Captions, and a transcript
c	Anyone with low vision, and everyone reading in sunlight	Higher contrast
d	Anyone slower: a motor impairment, a second language, a person interrupted	No time limit, or a warning and an extension
e	Anyone with limited fine motor control, keyboard users, most touchscreen users	A plain button as well
f	Blind users, and anyone whose images failed to load	Alt text
g	Deaf users, anyone without headphones, anyone who reads faster than the video talks	The same instructions in text
h	Anyone with a tremor, large fingers, or a moving bus	Bigger targets, more spacing

Most expensive to fix afterwards: e — a drag-only interaction is usually built into the structure of the page and replacing it means rebuilding the interaction. Accept **b** with reasoning: captioning a library of existing videos is genuinely costly, where captioning as you go is not.

Worth asking the class: which of these eight fixes would make the product uglier? Almost none of them, which undercuts the usual objection.

Task 7.2 · color alone

Good second clues: a shape as well as a color (check, triangle, cross) · a word next to the dot · a pattern such as solid, striped, hollow · position in a sorted list.

a. Shapes, words and patterns all survive black and white printing. A color that merely changes shade does not — and printing is exactly how these dashboards end up in meetings.

b. Black on white is easiest; mid-gray on white is harder; yellow on white is close to unreadable. What matters is the *difference in lightness* between the two, not whether the colors look pleasant together.

Task 7.3 · the alt text

a. 'Enrollments rose from about 800 in 2020 to 1,100 in 2026.' A chart's alt text should give the *finding*, not describe the axes.

b. 'The school's main gate on Rose Street, with the visitor entrance to the right.' Include what the reader needs to act.

c. Empty. This is the trap.

d. 'Error message: file too large.' Never leave text trapped in an image.

Why empty is better for c: a screen reader announces every image it meets. Describing a decorative swirl interrupts the reader with something that carries no meaning. Empty alt text tells the software to skip it entirely — which is different from having no alt text at all, where the reader falls back to reading out the filename.

Task 7.4 · keyboard only

a. Name, Email, Message, Send — the same order as they are read. Tab order should follow the visual order; when it does not, the page becomes impossible to follow.

b. The user is trapped. They cannot close it, cannot reach the page behind it, and often cannot escape without closing the whole browser and losing their work. This is called a keyboard trap and it is one of the worst failures on the list.

c. Because without a visible outline the user has no idea what pressing Enter will do. Designers remove that outline surprisingly often because they think it looks untidy.

d. Open it on click or keyboard focus as well as hover. Hover does not exist on a touchscreen either, so this helps far more people than intended.

Task 7.5 · the closing question

Designing for the hardest case produces something better for everybody, because the constraint forces clarity: captions, larger targets, high contrast and keyboard access are all used constantly by people with no impairment at all.

The same pattern appears everywhere — curb cuts for wheelchairs are used by prams, suitcases and delivery carts. Accept any answer reaching 'the edge case turns out to be the common case'.

Worth saying plainly: nobody has to be persuaded to use captions. They were simply better, once somebody was forced to build them.

LESSON 08 · TEACHER BRIEFING

The Algorithm Decided

READ BEFORE THE BELL

2 MINUTES

Students apply a scholarship scoring rubric to five real-shaped applicants, discover that one innocent criterion is quietly measuring something else entirely, then trace a recommendation loop closing in around a single interest.

You do not need to know this subject

Every score is simple addition and the totals are all in the key. The judgment questions have reasoning to look for rather than a single right answer.

Photocopy these pages only

Pages 42–44

One per student. **Do not photocopy this page or the answer key.**

What students need

A pen. Nothing else.

WHAT TO SAY TO START

> *"A rule written once and applied to five thousand people is a very different thing from a decision made five thousand times. Today you write one, apply it, and then find what you accidentally built into it."*

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–12	Read the rubric together	Do applicant A on the board with them
12–28	Task 8.1 — score and rank	Exact totals. Key has every one
28–40	Task 8.2 — find the proxy	Do not give this away. Let them find it
40–50	Task 8.3 — the recommendation loop	Simple to trace, unsettling to finish
50–55	Tasks 8.4 and 8.5, collect sheets	Discussion prompt below

Questions students will ask

"The rule treats everyone the same, so how is it unfair?"
That is the question the whole lesson exists to complicate. Task 8.2 answers it with arithmetic rather than argument.

"Is a computer even involved?"
Not in Task 8.1, deliberately. The point is that the unfairness is in the rule, not in the machine that runs it.

"Who wrote the rubric?"
Somebody who meant well. That is almost always the case, and it is why this is difficult.

If they finish early

Task 8.4 asks students to repair the rubric. Strong students should then be asked what their fixed version has lost, because every fix costs something.

If the computers are working

Students can compare recommendations on a fresh account with those on a well-used one, if that is practical, or simply describe how their own feeds have narrowed over a year.

Curriculum focus · for the regular teacher's records

Impacts of digital systems on society; algorithms in decision-making; bias, fairness and accountability; the rights of people affected by automated decisions. Discussion prompt for the last five minutes: *""The algorithm is not biased, it just uses the data." What work is that sentence doing?"*

LESSON 08 · THE ALGORITHM DECIDED

A rule applied five thousand times

UNDERSTAND

14 MIN

NAME

CLASS

DATE

The scholarship rubric

A school awards one scholarship. To be fair and consistent, the committee writes a points system and applies it to every applicant identically.

Points

Average mark — 80 or above: 3 · 70–79: 2 · 60–69: 1

Attendance — 95% or above: 2 · 90–94%: 1

Activities — 1 point each, maximum 2

Lives within 3 miles of school — 1 point

The applicants

	MARK	ATT.	ACTS	MILES
A	84	96%	2	2
B	78	91%	3	8
C	88	89%	0	1
D	72	97%	2	11
E	65	96%	1	3

Task 8.1 · Score them

CALCULATE

APPLICANT	MARK PTS	ATTENDANCE PTS	ACTIVITY PTS	DISTANCE PT	TOTAL	RANK
A						
B						
C						
D						
E						

a. Who wins? ______ **b.** Who came last? ______ **c.** Two applicants tie. Which? ______

d. The strongest student academically did not win. Explain why in one sentence.

______

LESSON 08 · THE ALGORITHM DECIDED

What the rule is really measuring

KEY SKILL

20 MIN

Task 8.2 · Find the proxy

JUDGE

A **proxy** is a criterion that looks like it measures one thing but reliably measures something else.

#	QUESTION	YOUR ANSWER
a	Why might the committee have included the distance point?	
b	What does living close to a school often depend on?	
c	Which two applicants lost that point, and what do they have in common?	
d	The rubric never mentions money. Is it measuring it anyway?	

e. The activities cap is capped at 2. Applicant B had three. What kind of student finds it easiest to have two organized activities? ______________________________

f. Remove the distance point and rescore. Does the winner change? __________ Does anything else?

____

Task 8.3 · The loop closes in

TRACE

A feed shows three videos. The rule: **whatever you watch, replace the two you ignored with more of that kind next time.** You start with Sport, Cooking and Space, and you happen to click Sport.

ROUND	WHAT YOU ARE SHOWN	WHAT YOU CLICK
1	Sport, Cooking, Space	Sport
2		Sport
3		Sport
4		—

a. After round 3, what has disappeared? ____________________

b. You now decide you would like some cooking videos. How would the system ever find that out?

____

c. The system reports that you love sport, and it has the click data to prove it. What is wrong with that claim?

____

Fixing it, and arguing about it

STRETCH

16 MIN

Task 8.4 · Rewrite the rubric

REDESIGN

a. Rewrite the scholarship rubric so it stops measuring the thing it should not. You may change, remove or add criteria.

b. What has your version *lost* that the original had? Every fix costs something.

c. An automated decision goes against someone. Name three things the system should be required to tell them.

1.
2.
3.

Task 8.5 · The last word

THINK

"The algorithm is not biased. It just uses the data." Explain in one or two sentences why that sentence is doing a great deal of quiet work.

Before you hand this in

CHECKPOINT

Check what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials

Answer key

The Algorithm Decided

NOT FOR STUDENTS

ANSWERS

Task 8.1 · the scores

	MARK	ATT	ACTS	DIST	TOTAL	RANK
A	3	2	2	1	**8**	1
B	2	1	2	0	**5**	3=
C	3	0	0	1	**4**	5
D	2	2	2	0	**6**	2
E	1	2	1	1	**5**	3=

a. A wins with 8. **b.** C comes last with 4. **c.** B and E tie on 5.

d. C has the highest average mark in the group and finishes last, because the rubric awards five of its eight available points for things other than academic achievement. Whether that is wrong depends entirely on what the scholarship is for — and nobody wrote that down.

Task 8.2 · the proxy

a. Probably good intentions: rewarding students likely to participate, or to arrive on time.

b. What a family can afford. Housing near a school is usually more expensive, so proximity tracks household income more closely than it tracks commitment.

c. B (8 miles) and D (11 miles). Both live further out, and both may well be traveling further *because* housing is cheaper there.

d. Yes. The rubric never says money and measures it anyway. That is what a proxy does, and it is why removing the sensitive field from a dataset does not remove the bias from the decision.

e. Students whose families can pay fees and provide transport, and who are not needed at home or working part time. A second proxy hiding in plain sight.

f. Without the distance point: A 7, D 6, B 5, E 4, C 3. The winner does not change, but **E drops below B** and C falls further. One point out of eight reordered a third of the field.

Task 8.3 · the loop

ROUND	SHOWN	CLICKED
1	Sport, Cooking, Space	Sport
2	Sport, Sport, Cooking	Sport
3	Sport, Sport, Sport	Sport
4	Sport, Sport, Sport	—

a. Space vanished after round 2 and cooking after round 3. Nothing announced it and nothing asked.

b. It cannot. You can only choose between what you are shown, and cooking is no longer among the options. The only escape is to search for it deliberately — which requires already knowing you want it.

c. The click data is real and the conclusion is still unsound. You clicked sport because sport was what you were offered. The system measured its own behavior and reported it as your preference — and every extra round makes the evidence look stronger.

This is worth naming: a feedback loop manufactures the data that justifies it.

Task 8.4 · repairing it

Good fixes: remove the distance point entirely · replace it with something that measures the intent directly, such as a written statement · award activity points for any commitment including paid work and caring for family · state what the scholarship is actually for and score against that.

b. What it loses: simplicity and consistency. A written statement has to be read and judged, which reintroduces exactly the human variation the points system existed to remove. That trade is real and there is no version without it.

c. The three things: **the decision and the reason for it** · **what data was used about them** · **how to have it reviewed by a person**. Accept ‘which rule was applied’ and ‘how to correct wrong data’.

Task 8.5 · the closing question

The sentence smuggles in three claims. It treats the data as though it fell from the sky, when somebody chose what to collect and what to leave out. It treats the criteria as neutral, when each one was selected by a person. And it moves responsibility onto a system that cannot hold any.

The honest version is: *this rule reproduces the patterns in the data we chose, at a scale no human decision could reach, and we are responsible for both halves of that.*

Accept any answer that puts a person back into the sentence.

LESSON 09 · TEACHER BRIEFING

The Smart Home Audit

READ BEFORE THE BELL

2 MINUTES

Students audit eight connected devices in one house, apply five questions nobody asks before buying, then follow how thousands of forgotten cameras take a website off the internet.

You do not need to know this subject

Everything is judgment and ordering. The key gives the reasoning for every device and the correct sequence for the attack.

Photocopy these pages only

Pages 47–49

One per student. **Do not photocopy this page or the answer key.**

What students need

A pen. Do not ask students to describe the devices in their own homes.

WHAT TO SAY TO START

> *“There is a computer in your doorbell. It has an operating system, a network connection and a password, and almost nobody has ever updated it. Today you find out what that means.”*

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–12	Read the panel together	Every smart device is a computer. Say it plainly
12–28	Task 9.1 — the audit	Expect argument about the TV. Argument is good
28–38	Task 9.2 — the five questions	Apply them to three devices
38–50	Task 9.3 — the botnet	Ordering, then the uncomfortable part
50–55	Tasks 9.4 and 9.5, collect sheets	Discussion prompt below

Questions students will ask

“Who would want to hack a doorbell?”
Nobody wants the doorbell. They want the network it sits on, or the doorbell’s ability to send traffic. Task 9.3 makes that concrete.

“How would you even know?”
Usually you would not. That is the most important sentence in the lesson.

“Should I not buy any of this?”
That is not the lesson. Task 9.4 asks for rules for buying well, not for refusing.

If they finish early

Task 9.4 asks what happens when the manufacturer shuts down. Strong students can then decide whether a device that stops working when a company closes was ever really sold to you.

If the computers are working

If the school permits it, the number of internet-connected devices visible from a public search engine for connected devices is startling. Keep it demonstrative and general; do not connect to anything.

Curriculum focus · for the regular teacher’s records

Data security and privacy in networked systems; risks of connected devices; evaluating the personal, social and ethical impacts of digital systems. Discussion prompt for the last five minutes: *“Your camera is watching your house. Who else is it watching for, and how would you find out?”*

Everything is a computer now

UNDERSTAND

14 MIN

NAME

CLASS

DATE

A doorbell with an operating system

Every connected device is a small computer with a network address, software that ages, and a password that usually still says `admin`.

The difference from a laptop is that nobody thinks of it as a computer, so nobody updates it, changes its password, or notices when it starts behaving strangely.

Task 9.1 · Audit the house

JUDGE

Rate each device 1 (little risk) to 5 (serious risk) and say what the worst realistic outcome is.

DEVICE	DETAIL	1-5	WORST REALISTIC OUTCOME
Smart TV	Has a microphone; last updated 2021		
Baby monitor	Password never changed from the default		
Doorbell camera	Footage stored on the maker's cloud		
Robot vacuum	Has mapped the floor plan of the house		
Smart light bulbs	No updates available at all		
Voice assistant	Always listening for its wake word		
Smart lock	Opens from a phone app		
Fridge	Connects to wifi to show the weather		

a. Which is the highest risk, and is it the one you expected? ______________________

b. Which device gains the *least* from being connected at all? ______________________

Five questions, and one attack

KEY SKILL
18 MIN

Task 9.2 · Before you connect anything

PROVE

The five questions: **Does it need the internet to do its job? Who sends updates, and for how long? What does it record? Where is that stored? What happens if the company closes?**

DEVICE	NEEDS INTERNET?	WHO UPDATES IT?	WHAT IT RECORDS	WHERE IT GOES
Doorbell camera				
Smart light bulb				
Voice assistant				

Which of the three fails the first question outright — it does not need the internet at all?

____

Task 9.3 · How a doorbell attacks a website

ORDER

Number these 1 to 6 to show how thousands of small devices are used to take a website offline.

ORDER	WHAT HAPPENS
	The website cannot answer real visitors and goes down
	A program scans the internet for devices still using default passwords
	The attacker sends one instruction to every infected device at once
	It logs in and installs its own software on each one
	Millions of requests arrive at one website in a few seconds
	The devices keep working normally so nobody investigates

a. Why does the owner of the doorbell never notice? ______________________

b. Whose fault is the outage: the attacker, the device owners, or the manufacturer? Defend your answer.

____

LESSON 09 · THE SMART HOME AUDIT

When the company disappears

STRETCH

16 MIN

Task 9.4 · Write the buying rules

DESIGN

A company that made smart plugs shuts down. The servers switch off. Every plug in every house stops responding to its app, permanently.

#	QUESTION	YOUR ANSWER
a	Why did the plugs stop, when the electricity did not?	
b	Which of the devices in Task 9.1 would survive this best?	
c	Did the buyers actually own the plugs?	

d. Write three rules you would follow before buying any connected device:

1. ______________________
2. ______________________
3. ______________________

Task 9.5 · The last word

THINK

A camera pointed at your front door is watching for you. Explain in one or two sentences who else it might be watching for, and how an ordinary owner could ever find out.

____

____

____

Before you hand this in

CHECKPOINT

Check what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

____

____

Teacher initials ______________

Answer key

The Smart Home Audit

NOT FOR STUDENTS

ANSWERS

Task 9.1 · the audit

DEVICE	RISK	WORST REALISTIC OUTCOME
Baby monitor	**5**	Live video and audio of a child, viewable by anyone who finds it. Default passwords are published lists, not secrets
Smart lock	**5**	The physical door opens. This is the only device on the list where the consequence is not digital at all
Smart TV	**4**	Five years of missed security updates on a device with a microphone, sitting inside the home network
Doorbell camera	**4**	Footage of everyone who visits, held by a company, sometimes shared with third parties or handed over on request
Voice assistant	**3**	Fragments of conversation recorded after a false wake-word and stored remotely
Robot vacuum	**3**	A room-by-room map of the house, which is worth more to a burglar than to a cleaner
Light bulbs	**2**	Unpatchable devices inside the network, useful as a foothold or as part of a botnet
Fridge	**2**	Same as the bulbs, with the added irritation of a fridge that needs software support

a. The baby monitor and the smart lock. Most classes expect the voice assistant, because it is the one that is talked about — the actual risks are the boring devices nobody thinks about.

b. The fridge. Showing the weather is not worth putting a permanently connected computer in a kitchen appliance that will still be running in fifteen years.

Task 9.2 · the five questions

Doorbell: needs internet only for remote viewing · updated by the maker, for an unstated period · records video and audio of the street · usually stored on the maker's servers.

Light bulb: does not need the internet at all — local control would do everything except switching a lamp on from a bus · often never updated · records when you are home, by inference · sends usage data to the maker.

Voice assistant: genuinely needs the internet, because the recognition happens on remote servers · updated regularly · records audio after the wake word · stored remotely, often reviewable.

The bulb fails the first question, and it is the one people buy most casually.

Task 9.3 · the botnet

1 A program scans for default passwords · **2** It logs in and installs its software · **3** The devices keep working normally · **4** The attacker sends one instruction to all of them · **5** Millions of requests hit one website · **6** The site goes down.

a. Because the doorbell still rings. It uses a little more data and a little more power, and nothing about its job changes. There is no screen to display a warning and no antivirus software running on it.

b. All three, and the reasoning is what earns the credit. The attacker committed the crime; the manufacturer shipped a device with a default password and no update mechanism; the owner never changed it. Increasingly, the law is moving toward blaming manufacturers, because they are the only party in a position to fix it at scale.

Task 9.4 · when the company goes

a. The app never controlled the plug directly. It talked to the company's servers, which talked to the plug. Remove the middle and no path is left, though both ends still work.

b. Anything that works locally — the vacuum still vacuums, the TV still shows channels, the lock still turns with a key if it has one.

c. They own the object and they were only ever renting the service that made it useful. That distinction is worth a full discussion.

d. Model rules: does it still work with no internet? · how long are updates promised, in writing? · can it be controlled locally? · change the password before it joins the network · put it on a guest network, away from the laptops.

Task 9.5 · the closing question

It is also watching for the company that stores the footage, anyone that company shares it with, anyone who gains access to the account, and anyone who compels its release — plus every neighbor, delivery driver and passer-by who never agreed to anything.

How to find out: read what the maker says it does with recordings, check who the account is shared with, look at whether footage leaves the house at all, and check whether the device offers local-only storage.

Accept any answer naming a party the owner did not choose.

LESSON 10 · TEACHER BRIEFING

Will a Machine Take This Job?

READ BEFORE THE BELL

2 MINUTES

Students break four jobs into their actual tasks and mark which ones a machine could do, check their predictions against what really happened to five earlier technologies, then rank the skills hardest to automate.

You do not need to know this subject

This is discussion and judgment. There is no correct opinion, and the key gives the reasoning to look for plus what actually happened in each historical case.

Photocopy these pages only

Pages 52–54

One per student. **Do not photocopy this page or the answer key.**

What students need

A pen. Nothing else.

WHAT TO SAY TO START

> *"The question is never whether a machine will take your job. It is which parts of your job, how quickly, and what is left over. Today you find out how to answer that for any job at all."*

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–12	Read the three tests together	Repetitive, rule-based, data-rich
12–28	Task 10.1 — break the jobs into tasks	The task level is the whole idea
28–40	Task 10.2 — predict, then check history	Do not reveal the key early
40–50	Task 10.3 — rank the skills	Expect disagreement. Ask for reasons
50–55	Tasks 10.4 and 10.5, collect sheets	Discussion prompt below

Questions students will ask

"Will there be any jobs left?"
Nobody knows, and anyone who says otherwise is guessing. What history shows is that jobs changed far more often than they vanished — which is Task 10.2.

"Is it worth learning to code if AI can code?"
Genuinely contested. Push them to argue it rather than answer it.

"Which job is safest?"
Task 10.3 is a better version of that question, because it asks about skills rather than job titles.

If they finish early

Task 10.4 asks students to invent a job that exists *because* of automation. Strong students can then name three jobs that exist today which could not have existed twenty years ago.

If the computers are working

Students can look up how many people work in a job today compared with twenty years ago. Bank tellers and typists produce genuinely surprising numbers in opposite directions.

Curriculum focus · for the regular teacher's records

Impacts of digital systems on society and work; automation and its economic and social consequences; evaluating the future implications of technology. Discussion prompt for the last five minutes: *"A machine can now do 40% of a job. Does the job disappear or change — and which is harder for the person doing it?"*

Jobs are made of tasks

UNDERSTAND

14 MIN

NAME

CLASS

DATE

The three tests

A task is easy to automate when it is **repetitive** (done the same way many times), **rule-based** (the right answer follows from the inputs), and **data-rich** (thousands of past examples exist).

Almost no *job* is all three. Almost every job contains *tasks* that are, which is why the useful question is asked one task at a time.

Task 10.1 · Break the job into tasks

ANALYZE

Mark each task **likely**, **partly** or **unlikely** to be automated, and give one reason.

JOB	A TASK IN IT	VERDICT	REASON
Radiologist	Scanning images for a known pattern		
	Telling a patient what the result means		
Truck driver	Driving on an empty highway		
	Reversing into a tight loading dock in the rain		
Teacher	Grading multiple choice tests		
	Working out why a quiet student stopped trying		
Accountant	Adding up and categorizing transactions		
	Advising a nervous small business owner		

What do all four **unlikely** tasks have in common? ____________________

LESSON 10 · WILL A MACHINE TAKE THIS JOB?

Check it against what happened

KEY SKILL

18 MIN

Task 10.2 · Predict, then look

PREDICT

Write your prediction first. Your teacher has what actually happened.

#	THE TECHNOLOGY	YOUR PREDICTION: WHAT HAPPENED TO THOSE JOBS?
a	ATMs arrived in banks	
b	Spreadsheets replaced hand-kept ledgers	
c	Self-checkouts arrived in supermarkets	
d	Autopilot became standard in aircraft	
e	Spell-checkers arrived in word processors	

a. Which of your five predictions were wrong? ____________________

b. In which cases did the number of workers *rise*? ____________________

Task 10.3 · Rank the skills

RANK

Number these 1 (hardest to automate) to 8 (easiest).

SKILL	1-8	SKILL	1-8
Persuading a reluctant person		Deciding what problem is worth solving	
Copying figures between systems		Translating routine text	
Repairing something in a cramped space		Comforting someone who is upset	
Summarizing a long document		Spotting a pattern in a million records	

a. Justify your number 1: ______________________________

b. Two of these eight moved a long way down the list in the last five years. Which two?

____

What the job becomes

STRETCH
16 MIN

Task 10.4 · Invent the job

EXTENSION

Write a short job description for a role that exists **because** of automation, not despite it. It must be a real kind of work, not a joke.

JOB TITLE	

What they do all day:

a. Which automated system does this job exist to support, check or repair?

b. Could this job have existed twenty years ago? ____________ Why? __________________________

Task 10.5 · The last word

THINK

A machine can now do 40% of somebody's job. Does the job disappear or change? Say which you think is harder for the person doing it, and why.

Before you hand this in

CHECKPOINT

Check what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials __________________________

Answer key

Will a Machine Take This Job?

NOT FOR STUDENTS

ANSWERS

Task 10.1 · task by task

TASK	VERDICT	WHY
Scanning images for a pattern	**Likely**	Repetitive, rule-like, and millions of labeled examples exist. Already in use
Telling a patient what it means	**Unlikely**	Requires judgment about a person, and accountability for the conversation
Highway driving	**Likely**	Predictable environment, clear rules, enormous data
Reversing into a dock in the rain	**Partly**	Unpredictable, physical, and expensive to get wrong
Grading multiple choice	**Likely**	Already automated everywhere
Why a quiet student stopped trying	**Unlikely**	Needs relationship, context and things nobody wrote down
Categorising transactions	**Likely**	Rule-based and data-rich
Advising a nervous owner	**Unlikely**	Judgment, trust, and responsibility for the advice

The common thread: all four unlikely tasks involve a person, an unpredictable situation, and somebody being accountable for the outcome. Notice also that in every job the automatable task is the one people describe as the ‘real work’ — and the one left over is the part nobody was trained for.

Task 10.2 · what actually happened

a. ATMs: the number of bank tellers **rose** for two decades. Branches became cheap to run, so banks opened more of them, and tellers moved from counting cash to selling services. Almost every class predicts the opposite.

b. Spreadsheets: hundreds of thousands of bookkeeping jobs disappeared, while the number of accountants and financial analysts **grew**. The tedious part vanished and the advisory part expanded.

c. Self-checkouts: fewer cashiers per store, but new roles supervising, assisting and handling theft. A genuine net reduction, unlike the first two.

d. Autopilot: flight crews shrank from five to two, but pilots were never removed. The work became monitoring, and a new problem appeared: staying alert while a machine does the flying.

e. Spell-checkers: nobody lost a job. Editors still exist and now work on things a spell-checker cannot judge.

Task 10.3 · the ranking

A defensible order, hardest to easiest: **1** comforting someone upset · **2** deciding what problem is worth solving · **3** persuading a reluctant person · **4** repairing something in a cramped space · **5** summarizing a long document · **6** translating routine text · **7** spotting a pattern in a million records · **8** copying figures between systems.

Accept any order argued well. The reasoning matters far more than the sequence.

b. Summarizing and **translating** both moved dramatically toward the easy end within a few years, and both were confidently placed near the hard end a decade ago. That is the most honest thing in the lesson: the list moves, and it moves faster than anyone predicts.

Note that physical repair in an awkward space remains hard, which surprises students who assume physical work is the vulnerable kind.

Task 10.4 · jobs created by automation

Good answers describe real categories of work: training and correcting a model · auditing automated decisions for fairness · maintaining and repairing the machines · handling the cases the system refused or flagged · explaining an automated decision to the person affected · deciding what a system should and should not be allowed to do.

The strongest answers notice the shape: automation creates work at the **edges** — setting up, checking, repairing, and dealing with everything that does not fit.

b. Most could not have existed twenty years ago, because the system they support did not. That is precisely how new work has appeared after every previous wave.

Task 10.5 · the closing question

Usually the job changes rather than disappears — and the change is often harder to live through than a clean ending. The tasks removed are frequently the ones the person was good at and enjoyed, leaving the parts nobody trained them for and often more of them.

It also changes who is hired next: a role redesigned around the remaining 60% may require different skills from the one that was advertised.

Accept either answer. The credit is for recognizing that ‘the job survived’ is not the same as ‘the person was fine’.

AFTER THE LESSON

Handover note

LEAVE THIS ON THE DESK

90 SECONDS

Photocopy this page or fill it in directly. Ninety seconds of your time saves the regular teacher a great deal of guesswork.

SUBSTITUTE TEACHER	CLASS	DATE	PERIOD

Lesson used

CHECK ONE

- ☐ 01 · Pictures Made of Numbers
- ☐ 02 · Sound Made of Numbers
- ☐ 03 · Stacks, Queues and the Undo Button
- ☐ 04 · Reading the Logs
- ☐ 05 · Version Control by Hand
- ☐ 06 · Building a School Network
- ☐ 07 · Design for Everyone
- ☐ 08 · The Algorithm Decided
- ☐ 09 · The Smart Home Audit
- ☐ 10 · Will a Machine Take This Job?

How far did the class get?

- ☐ Everyone finished the main tasks
- ☐ Most finished; some are partway
- ☐ We got through about half
- ☐ We did not get far — see notes

Sheets

- ☐ Collected and left on the desk
- ☐ Students kept them
- ☐ Graded against the answer key

Anything the regular teacher should know

NOTES

Students to mention · for any reason, good or otherwise

NAMES

Thank you

Walking into an unfamiliar subject is not easy. Leaving a clear note behind is the part that makes the next teacher’s day work, and it is genuinely appreciated.

THE FINE PRINT

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DIGITAL VECTOR

VOLUME FOUR

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Curriculum alignment

CSTA K–12 Computer Science Standards, levels 2 and 3A. Across the ten lessons this volume touches:

- Representing images: pixels, resolution and color depth
- Representing sound: sampling, bit depth and file size
- Abstract data structures: stacks and queues
- Interpreting logs, evidence and audit trails
- Version control and collaborative development
- Network devices, topologies and capacity planning
- Accessibility and inclusive design
- Automated decision-making, fairness and accountability
- Security and privacy of connected devices
- Automation and the future of work

Individual lesson teacher pages name the specific focus for that lesson.

The rest of the series

Volume 1 · Foundations

Algorithms, binary, trace tables, flowcharts, cyber attacks, networks, sorting, debugging, AI and digital footprints.

Volume 2 · Data, Systems and Design

Data cleaning, misleading charts, databases, input-process-output, compression, encryption, interface design, functions, project planning and the cost of free apps.

Volume 3 · Machines and Meaning

Lists, Boolean logic, hardware, files, requirements, machine learning, DNS, search ranking, control systems and copyright.

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Physical Computing with the Raspberry Pi 5 · AI Literacy for Secondary Students · and more.

Found an error, or want a lesson on something else?

Feedback is genuinely read. If a lesson did not land with your class, or you need a sub plan on a topic that is not here, get in touch through digitalvector.com.au and it will be considered for the next update.

EMERGENCY SUB PLANS PACK

Digital Technologies · Grades 7–10

Volume Four · ten lessons · no preparation · no computers required